House Price Prediction Using NumPy

Task 1 - AI-ML Mini Project

Name: Sakshit Shivam

1. What is the project about?

In this project, I used house sizes and their prices to make a simple program that predicts the price of a house based on its size.

I used NumPy for the calculations instead of using a machine-learning library. This helped me understand the basic working of linear regression.

The prices in the dataset are given in lakhs.

2. Dataset and NumPy

House size (sq ft)	Price (lakhs)
500	15
800	24
1000	30
1200	36
1500	45
1800	54
2000	60
2200	66
2500	75
2800	84

I also calculated the mean, median, minimum, maximum and standard deviation to get a basic idea of the data.

3. Model and testing

I used the first 8 houses for training and the last 2 houses for testing.

The model uses a straight-line equation:

Predicted price = slope x house size + intercept

Calculated slope: 0.03 lakh per square foot
Calculated intercept: 0.0
Prediction for a 1300 sq ft house: Rs 39 lakh

4. Results

Actual test prices: 75 and 84 lakhs
Predicted test prices: 75 and 84 lakhs
Mean Absolute Error (MAE): 0
Mean Squared Error (MSE): 0

The model got both test predictions correct for this dataset. The dataset follows a very simple linear pattern, so these results should not be expected with every real-world dataset.

5. Extra challenges

Outlier check: I added a price of 200 lakhs and checked how the mean and median changed. The mean changed from 48.9 to about 62.64.

Baseline comparison: I compared the model with a simple method that always predicts the average training price.

Baseline MAE: 38.25 lakhs
Linear regression MAE: 0 lakhs

6. What I learned

This project gave me practice with NumPy arrays, basic statistics, training and testing data, and linear regression.

One limitation is that the dataset is small and only uses house size. Real house prices can also depend on location, condition and other factors.

Thank you